

- 4 (a) (i) The internal energy of a system is the sum of the kinetic energy due to the random motion of its molecules, and the potential energy due to intermolecular forces of attraction.

Note: The majority omitted the fact that the kinetic energy is associated with the random motion of the atoms/molecules.

- (ii) The first law of thermodynamics states that the increase in internal energy of a system is the sum of the thermal energy added to the system by heating and the work done on the system.

(b) (i) Work done $W = p\Delta V = (1 \times 10^5)(20 - 5) \times 10^{-6} = 1.5 \text{ J}$

- (ii) The completed table is shown below.

section of cycle	heating supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
$A \rightarrow B$	zero	4.2	4.2
$B \rightarrow C$	- 8.5	zero	- 8.5
$C \rightarrow A$	5.8	- 1.5	4.3
Net	- 2.7	2.7	zero

The last row 'Net' is included to enable the completion of the table.

- 5 (a) (i) The electric field strength at a point in an electric field is the electric force per unit